

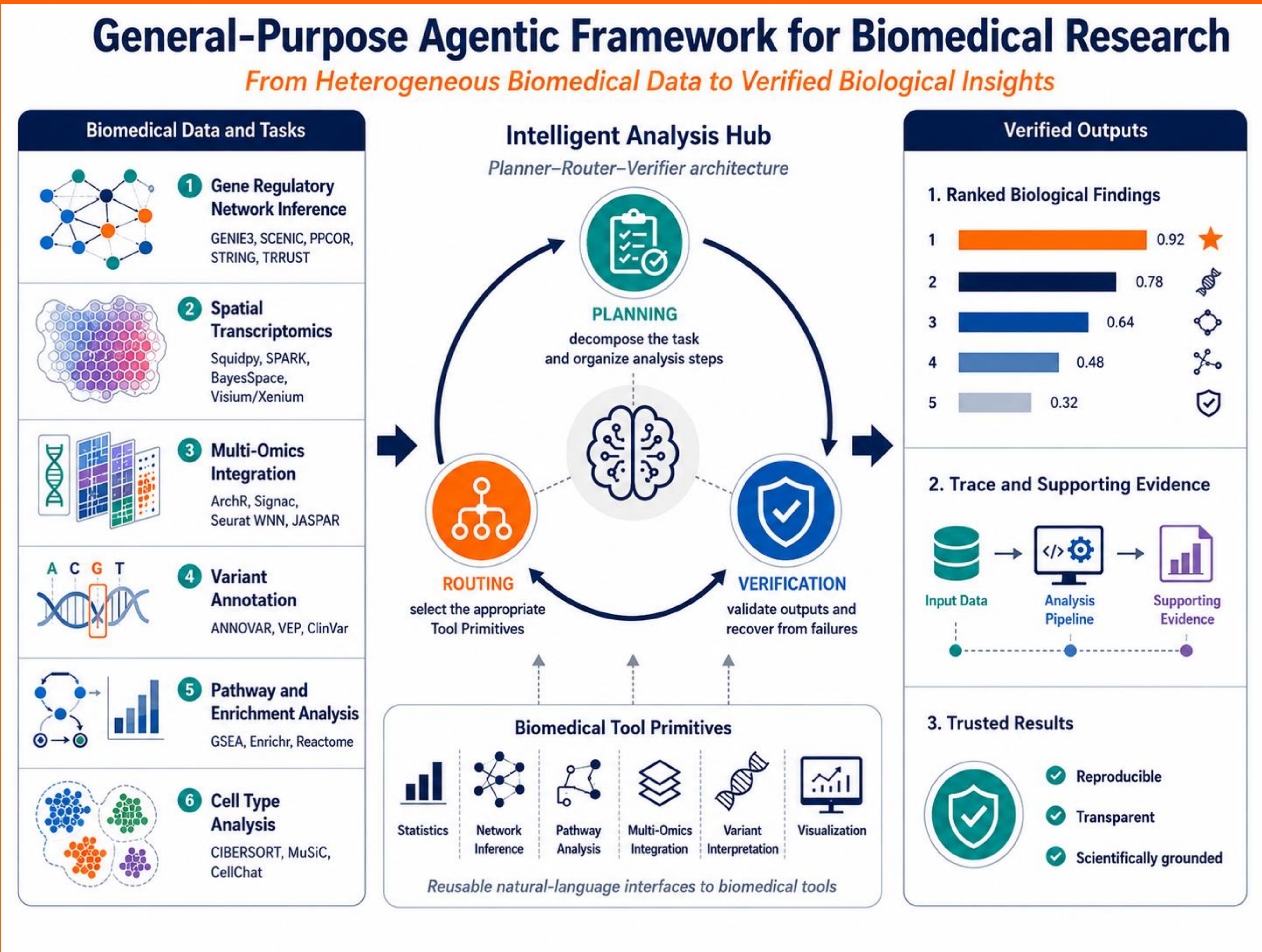
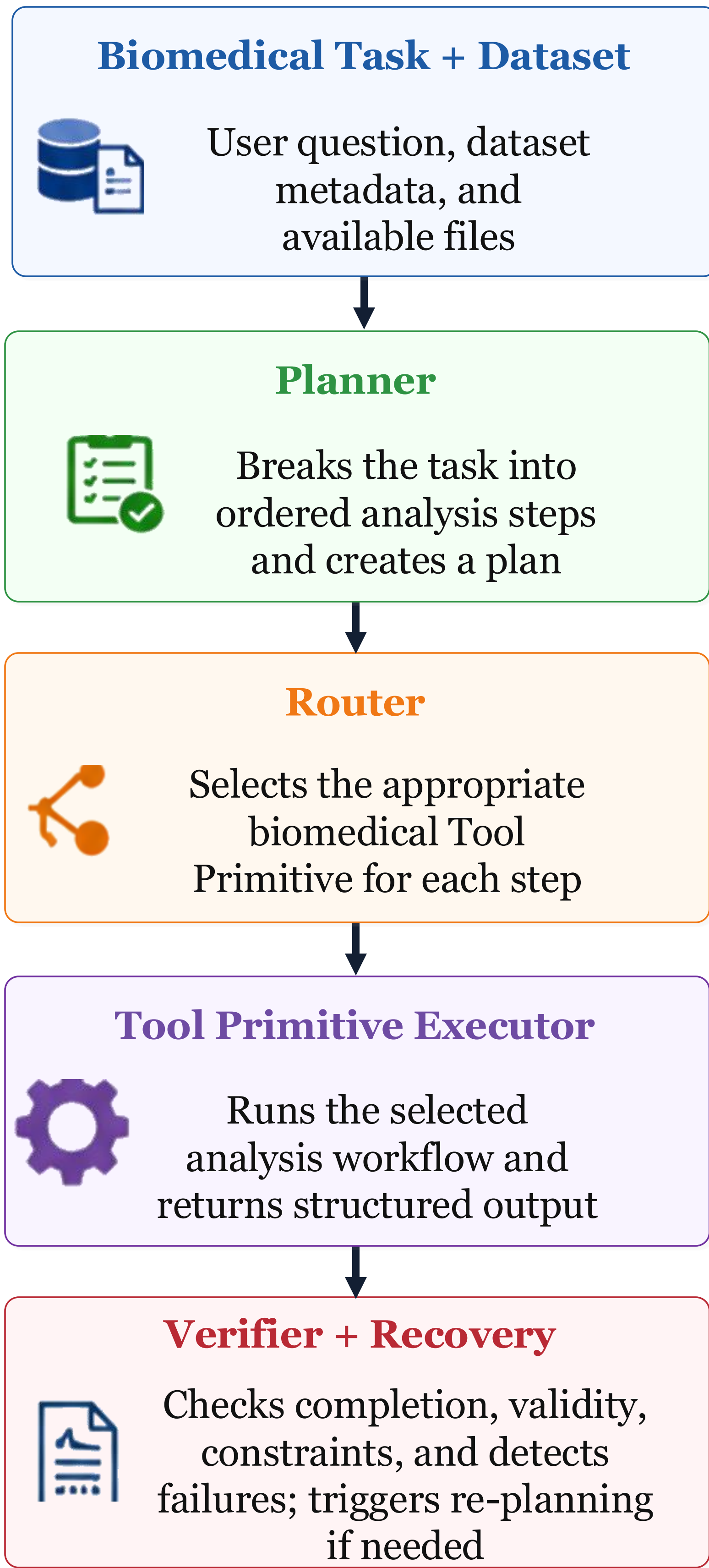
General-Purpose Agentic Framework for Biomedical Research

Parth Agarwal, Dr. Wang
School of Information Sciences, University of Illinois Urbana-Champaign

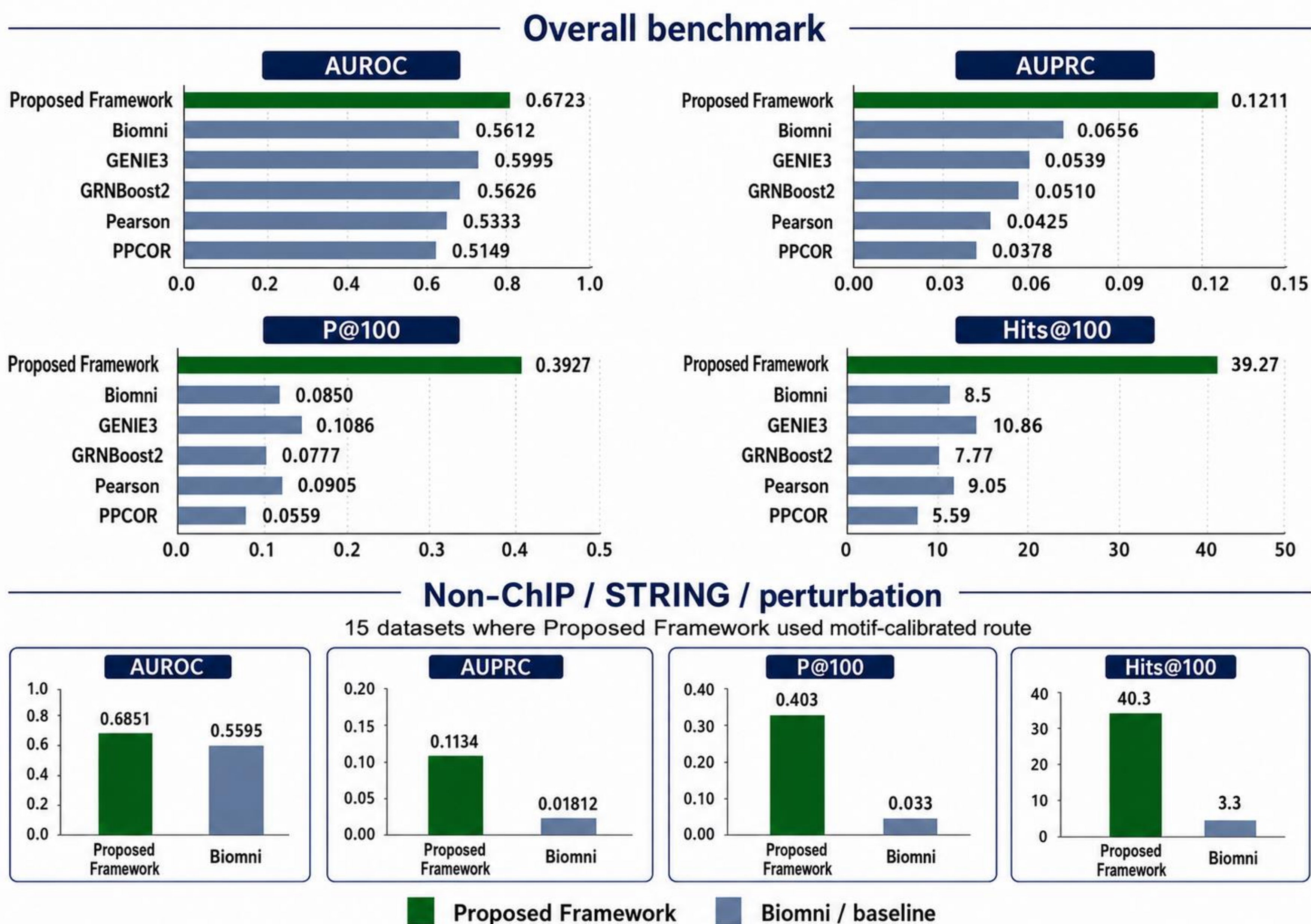
INTRODUCTION

- Biomedical research increasingly depends on complex data-analysis workflows across gene expression, regulatory genomics, single-cell, and multi-omics datasets.
- Current tool-augmented AI agents often focus on selecting tools, but scientific reliability also requires correct execution, verification, and recovery from failures.
- This project explores a planner-router-verifier framework that wraps biomedical tools as reusable Tool Primitives to support structured, traceable, and generalizable biomedical analysis.

Planner-Router-Verifier Workflow

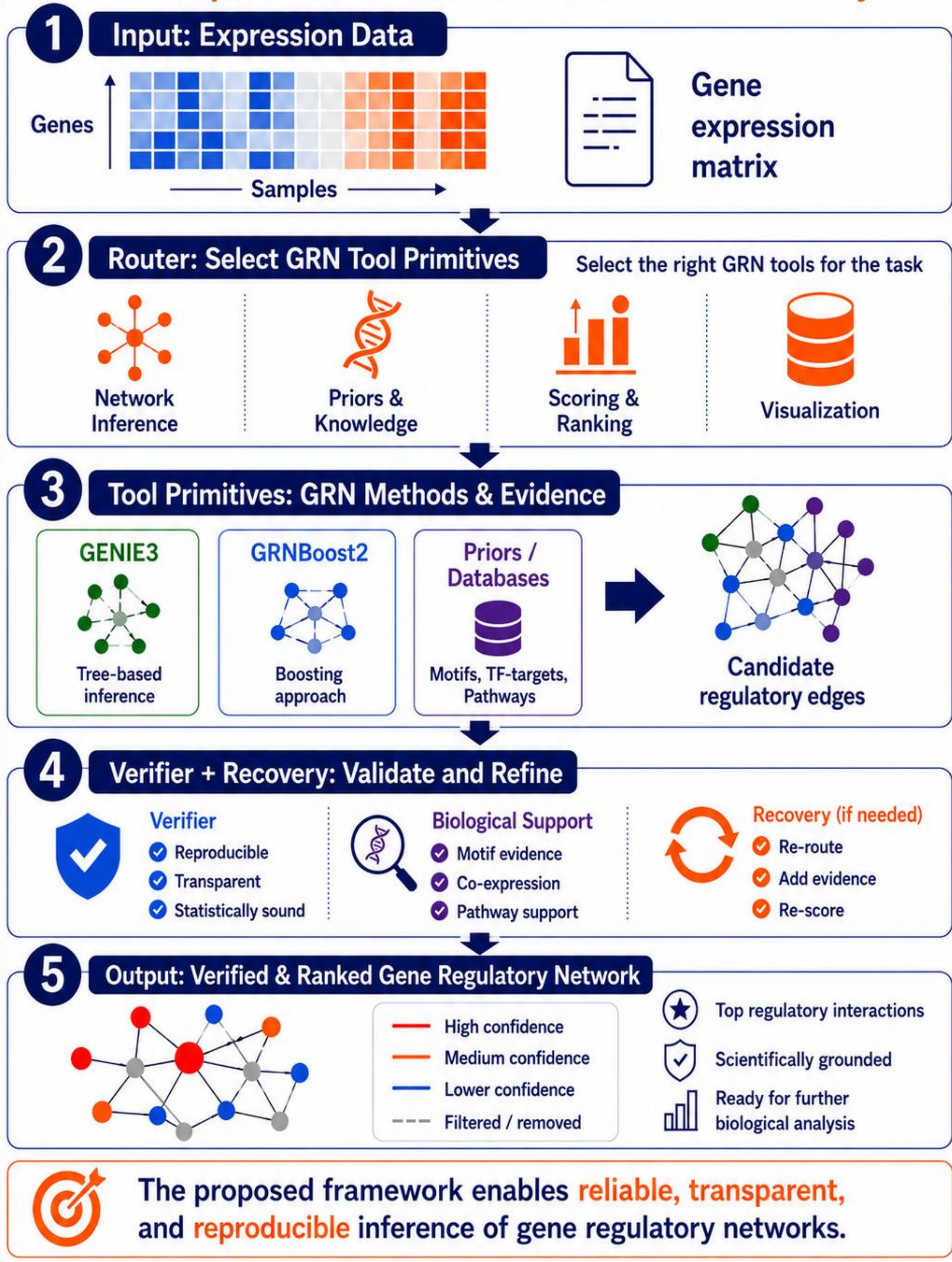


RESULTS



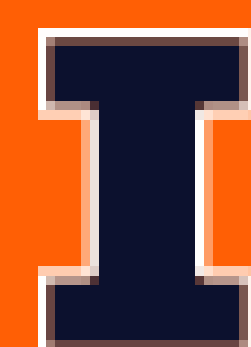
From Data to Verified Biological Edges

Proposed Framework in Action: GRN Case Study



ACKNOWLEDGEMENTS

I sincerely thank Dr. Wang for giving me the opportunity to participate in this research and for his guidance, support, and encouragement throughout this work and its presentation at the conference.



School of
Information Sciences